this and static

Agenda

1. Concept of this pointer
2. Use of this pointer
3. Static
    - Data
    - Function

1. Concept of this pointer

- There can be many objects that use single set of member functions
- this pointer connects invoking object with a member function
- Data type of this pointer
- this pointer is a local variable. Created on stack when function is called. Removed from stack when the function returns
- It is a hidden parameter added by compiler for every non-static function at the start of parameter list

2. Use of this reference

- To avoid shadowing of data members by local variables of same name
- Return invoking object from a member function

3. Static keyword in Java

- Static applicable for
  - Variable
  - Function

3. Static variable

- Non-static data created separately for each object.
- Static data created only once and shared by all objects that that class
- Created on Data segments of program. Neither on stack nor on heap.
- Must be created with explicit global declaration with class name
- Value of static data can be changed. Once changed, the new value is visible to all objects

- Examples of:
  - Non-static: Name, Id, Fees, Marks of student
  - Static: University, Course name, Course id

  - Other examples

3. Static function

- It is member function of class
- Called without object. Using class name.
- No this pointer in static function
- Can’t access non-static data
- Uses for managing static data